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A Novel Approach for Text Extraction and Word Segmentation from Handwritten Document Images Using CNN-RNN Technique

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Abstract—Optical Character Recognition is a technology that takes an optical image of a character as input and generates the corresponding character as output. Its applications span a wide array, encompassing fields such as traffic surveillance, robotics, and the digitization of printed material. Implementation of Optical Character Recognition often involves Convolutional Neural Networks, a widely adopted architecture within the realm of deep learning. Traditional Convolutional Neural Network classifiers excel in learning crucial 2D features within images and subsequently classifying them. This classification process is typically carried out utilizing a SoftMax layer. In this paper, the authors described the optical character recognition by using refined versions of Convolutional Neural Networks and a Recurrent Neural Network classifier. The quality of text recognition was assessed using Character Error Rate and Word Error Rate. Two datasets, IAM and RIMES, were utilized, each divided into training and testing subsets. Accuracy, precision, and recall were calculated based on these divisions. The experimental findings revealed that the Convolutional Neural Networks method achieved notably higher accuracy rates across both datasets, reaching 89.3% and 86%, respectively.

Index Terms— CNN (Convolutional Neural Network), Feature Extraction, Optical Character Recognition (OCR), RNN, Text Extraction.

I. INTRODUCTION

OVER the past two decades, there has been significant progress in the analysis of printed documents, particularly in the field of Devanagari script and Hindi-based processing systems [1]. These advancements have led to the establishment of reliable methods for searching, extracting, and retrieving information from both text and image-based documents [2]. Printed text processing methods are consistently more straightforward than handwritten ones [3].

The challenges associated with reading handwritten text mostly arise from the use of cursive writing, the presence of overlapping and touching letters, and the changes in height, size, and spacing between characters and sentences [4]. The textual material included in photographs contains important and vital information that assists in fully understanding the subject of the image. Extracted text from images is essential.

In several applications, including document analysis, automobile plate recognition, video content analysis, document retrieval, and aiding those with visual impairments [5] [6].

In the current era of a digital environment, where information is transmitted through networks, there is a need for mechanisms that can prioritize the development of highly accurate, efficient, and effective techniques for text recognition and detection. Additionally, there is a need for methodologies that can locate, identify, and extract text from documents that contain both machine-printed and handwritten images. A document image has several components, including textual content, visual representations, and graphical features, such as line drawings and sketches. These documents are generated by scanning a range of sources, including journals, historical document images, deteriorating document photos, handwritten text images, printed document images, multi-color book covers, and newspapers [7] [8].

The broadened scope of text document image processing includes many crucial phases, such as preprocessing, character recognition, word recognition, and document classification. Now, several researchers are dedicating their focus to it [9]. Several Natural Language Processing (NLP) and image processing applications have been built using Indian text sources that are either monolingual or multilingual [10][11]. Therefore, NLP and image processing are vast fields that have a great deal of applications that are based on bilingual processing. Text processing systems that are based on NLP include solutions for applications that operate on plain text documents, while text processing systems that are based on image processing include

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solutions for applications that need to be scanned printed, and handwritten documents [12] [13].

The process of transforming printed text into editable text may provide significant assistance to companies and organizations [14]. These pictures may include data in any language, exhibit varying levels of quality, be either black and white or colorful, and may incorporate metadata or other images. Hence, the scope of extracting text from printed photographs goes beyond the constraints of OCR technology [15].

Scanned documents sometimes provide many issues, such as poor contrast, low resolution, color bleeding, complicated backdrop, and unknown text characteristics such as color, size, location, orientation, and layout, as seen in Figure 1. Text extraction becomes challenging and illegible when dealing with multi-color document images, since the foreground text becomes blended with the background owing to color mixing, as seen in Figure 2. OCR systems of high caliber exhibit subpar performance when confronted with challenges such as low resolution, intricate backgrounds, blurriness, diverse layouts, and pictures including stray markings [16].

Typically, OCR works best when applied to photographs with uncluttered backgrounds. This introduction offers a concise summary of the difficulties and approaches associated with extracting text and information from handwritten document pictures. The main emphasis is on the importance of this field in facilitating the effective digitization and analysis of handwritten documents for a range of practical uses. Researchers and practitioners are using image processing, pattern recognition, and artificial intelligence to develop reliable and scalable methods for extracting important text and information from handwritten document images. This progress is helping to advance technologies for document analysis, archival, and information retrieval.

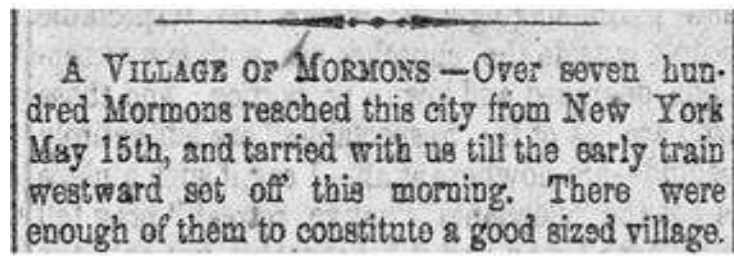


Fig. 1. A fragment from the grayscale scan of document BM_24May1859_p2_c3 from the 19th Century Mormon Newspaper Article collection.[37].



Fig. 2. Multi-color book cover image

It involves developing a system or algorithm that can accurately and efficiently convert handwritten text from document images into digital text using OCR technology. This task typically includes addressing challenges such as variations in handwriting styles, noise in the images, different languages, and formatting issues. The objective is to create a robust and reliable solution that can automatically extract text from handwritten documents to facilitate tasks such as digitizing archives, automating data entry, and enabling text searchability in scanned documents.

The paper's organization can be summarized as follows. Section II offers an overview of a review of the literature, discussing various methods for Automated Extraction of Text from Handwritten Document Images. Section III offers a comprehensive explanation of our proposed data normalization process. Moving forward to Section IV, the dataset description and system configuration are explained, and a summary of the experiments conducted to assess and validate the proposed approach is presented in Section V. Finally, in Section VI, the paper is concluded.

II. REVIEW OF LITERATURE

This section describes various authors' previous studies based on automated text extraction from handwritten document images.

Handwritten document recognition using an end-to-end architecture free of segmentation and also makes use of the Document Attention Network to label layouts and recognize text [1]. The system for classifying printed documents in Hindi that combines fuzzy logic and support vector machines (SVM) is presented in this research. To effectively categorize printed Hindi papers, the system makes use of a bi-level classifier and a tri-layered segmentation technique [2]. The research introduces a deep learning-based object detection technique called Faster R-CNN for text line segmentation of ancient Indian handwritten texts. The suggested method successfully segments text lines in difficult old documents with different script kinds (Devanagari and Maithili) and skew [3].

The system provides a Convolutional Autoencoder (CAE)-based approach for compressing handwritten Devanagari text documents. It has been demonstrated that the CAE methodology performs better in terms of compression efficiency and image quality than conventional approaches such as JPEG, albeit at the cost of greater computer resources [5]. In order to effectively categorize printed and handwritten text documents, Puri and Singh's study "A fuzzy matching based image classification system for printed and handwritten text documents" provides a hybrid image classification system that blends SVM and fuzzy logic [9].

This work opens up new possibilities for handwritten text analysis and understanding without transcription by showcasing the potential of semantic word representations for recognition-free NLP on handwritten texts [10]. In order to increase word segmentation and pre-processing accuracy, the paper incorporates NLP approaches. This entails using syntactic and semantic data to interpret and analyze the handwritten text more effectively [11].

With applications in document archiving, retrieval, and automated processing, his work offers cutting-edge methods for effectively analyzing and processing bilingual document pictures. It is also possible to expand the approaches to handle multilingual documents [12].

The study advances handwritten text analysis and processing methods by presenting an efficient deep learning-based method for text identification in handwritten documents [13]. The system advances the state-of-the-art in this field without requiring explicit text transcription by presenting a novel method that uses transformer networks to address named entity extraction and handwriting recognition from historical document images in tandem [14].

This work provides a low-tech hybrid OCR solution that effectively improves handwritten text recognition performance by utilizing the strengths of various recognition methods. This development advances continuing attempts to improve the usability and accessibility of handwritten text processing [15]. Work offers insightful information about the capabilities and constraints of OCR engines for text recognition in a variety of real-world document settings [16]. The use of graph neural networks to successfully capture the spatial and semantic links between various parts in handwritten writings allows for robust information extraction [17].

The principal contributions are: 1. The method recognizes and localizes text from natural scene photos using SVM-based techniques. 2. The suggested Text Information Extraction (TIE) system excels in extracting text from complicated natural scene photos, surpassing traditional OCR algorithms [18]. The method accurately assesses the quality of handwritten photo document images by taking into account elements such as text clarity, background noise, and geometric distortions [19].

The technique uses instance segmentation, which implicitly defines lines during training, to segment text lines in handwritten texts successfully. Tests conducted on reference datasets show that the suggested strategy outperforms conventional techniques for text line extraction [20]. The approach combines a graph-based representation of the spatial and semantic links between text items with image processing techniques to predict the shape of tables in bills. This makes it possible to extract structured data from intricate invoice layouts with robustness [21].

Using a combination of image processing techniques and machine learning models, the system efficiently separates text lines in handwritten papers with struck-out content. This makes it possible to reliably retrieve text even when there are deleted or crossed-out passages [22].

Text is extracted from the document image using a deep neural network, and the background is recreated using a Gaussian mixture model. The authors suggest a second method that uses deep neural networks to accomplish both background reconstruction and foreground extraction (including text with its original color), eliminating the necessity for human parameter setting in Gaussian mixture modeling [23].

This technique makes use of the distance transform to handle complex layouts, overlapping text, and different skew angles while efficiently segmenting text lines in handwritten texts. Instead of employing numerous features or intricate ensemble models, the writers present a writer identification method that is based on a single feature extraction strategy. As a result, the method is more effective and simpler to use in practical settings [24].

The authors treat the task of text-line extraction as an image-to-image translation problem and apply GANs to it. For the generator and discriminator, they employ the U-Net architecture with various combinations of loss functions (GAN loss, L1 loss, and L2 loss) [26]. The authors hope to aid academics and practitioners in comprehending the inner workings of these potent neural networks by offering a succinct and straightforward description of the fundamental CNN building blocks [28].

A two-stage CNN-RNN-based architecture for license plate recognition is proposed by the authors. In the first step, a CNN is used to locate and identify the license plate area in an input image. After that, an RNN is used in the second stage to identify the characters on the localized license plate. [34] Using a variety of AI techniques—such as CNNs, RNNs, SVMs, KNN, and LSTM—together offers a reliable and effective way to improve handwritten text recognition system performance [33]. Provides an attention-based model that can recognize handwritten paragraphs from beginning to end without the need

to first partition the input. The model automatically locates and transcribes the text by integrating both covert and overt attention techniques [36].

In the field of document analysis and recognition, the capability to execute end-to-end handwritten paragraph recognition without requiring previous line segmentation is a noteworthy breakthrough. This method may increase the handwriting recognition systems' resilience and effectiveness, particularly in real-world situations when the input papers may have different layouts and text distributions [37].

A. Comparison table

Table I below presents the authors' study strategies and provides an overview of the pertinent literature.

Table I. Comparison table of techniques used for word segmentation

Reference	Year	Techniques Used	Datasets	Outcomes
[1]	2023	Fully CNN	RIMES READ 2016	The DAN model achieved a single-page character error rate (CER) of 4.54% on the RIMES dataset. The DAN model produced CERs of 3.43% and 3.70% at the single-page and double-page levels, respectively, on the READ 2016 dataset.
[2]	2019	SVM and Fuzzy matching	Hindi printed images.	For training and testing, the average execution times for SL character recognition were 0.20375 and 0.22675 seconds, respectively. For training, the accuracy of character identification was 74.61%, while for testing, it was 80.73%.
[3]	2023	Faster Region-Convolution Neural Network (Faster R-CNN)	Custom dataset of ancient Devanagari handwritten documents	On the test set, the suggested Faster R-CNN-based approach produced a text line segmentation accuracy of 96.41%.
[5]	2013	Convolutional Autoencoder (CAE)	Handwritten Devanagari document images	According to the authors, the CAE model produces significantly superior compression and decompression outcomes than traditional image compression methods like JPEG.
[9]	2020	Bi-leveled image classification, SVM	Hindi printed images	The SVM-based classifier's accuracy in character recognition was 80.73% during testing and 74.61% during training.
[10]	2023	HTR-based model that combines handwritten text recognition (HTR) with a task-specific NLP system	IAM Dataset, Synth12K	The findings demonstrate the promise of recognition-free natural language processing (NLP) on handwritten document images in resolving the error propagation issue found in HTR-based systems.
[11]	2022	Deep learning and natural language processing (NLP)	Handwritten Tamil word images	When compared to conventional feature extraction techniques, the suggested method recognized handwritten Tamil words with higher detection accuracy.
[13]	2022	CNN	IAM Dataset	On the IAM dataset, the suggested CNN-based text identification model yielded an accuracy of 94.7%.

[14]	2022	Self-attention transformer encoder-decoder architecture	Esposalles Database (historical documents database)	For the entire job of combining handwriting recognition and named entity recognition on the Esposalles dataset, the suggested transformer-based model obtained state-of-the-art performance.
[15]	2022	OCR	IAM Dataset	On the IAM dataset, the suggested combination of OCR algorithms produced a character recognition accuracy of 92.5%.
[16]	2023	OCR	Custom dataset for vehicle license plate, SROIE dataset for receipts, IAM dataset	The Google Cloud Vision API and Microsoft Computer Vision API worked best on photos of car number plates, obtaining accuracy rates above 90%. With accuracy rates of about 85%, the Amazon Textract and Abbyy FineReader Engine fared better than the other OCR engines for receipt text recognition. With an accuracy of 80% on the handwritten text dataset, the Tesseract OCR engine outperformed the Google Cloud Vision API.
[17]	2024	GCN	IAM dataset	It is possible to expand the suggested method to additional sequence-to-sequence applications because of its generalizability.
[18]	2024	SVM	ICDAR 2003	The experimental outcomes yielded favorable values across all performance metrics in comparison to the established methodologies.
[19]	2023	“Classification-Based Transfer Learning	DIQA	The performance of each model is significantly influenced by various hyperparameters, leading to differences in results that may vary from a few points to a considerable percentage.
[20]	2023	DL	ICDAR2017, ICDAR2015-HTR, and Bozen	The average pixel IU and line IU scores across 9 datasets with diverse handwritten scripts, layouts, page backgrounds, line orientations, and interline spacings are 0.858 and 0.899, respectively, when the model is trained end-to-end without dataset-specific pre- or post-processing.
[21]	2022	Supervised Learning	English data and Czech data	In addition, OCRMiner has the capability to determine with a 98% degree of precision whether or not a picture is on the first page of an invoice image.
[22]	2022	DenseNet Deep Learning model	ICDAR2013 dataset and ICDAR2019 HDRC dataset	Experimental findings on two standard databases, a new problem-focused database including cases of struck-out text in handwritten documents, and the state-of-the-art demonstrate the effectiveness and utility of the suggested steps, as well as their better efficiency in comparison with current approaches.
[23]	2021	Deep Learning	H-DIBCO 2018 and DIBCO 2017	It is possible to recreate the (anticipated) original document picture by combining the repaired foreground and background photos. Even with significant deterioration, our approaches provide aesthetically acceptable findings for photographs of handwritten documents.
[24]	2021	TLE	DARPA-MADCAT, HIT-MW, ICDAR 2009, and ICDAR 2013	Both the experimental findings and the similar analysis conducted on the six datasets described earlier demonstrate that our strategy consistently achieves better outcomes than the state-of-the-art methods.

[25]	2020	SVM	IAM Dataset	In identifying the writers in the IAM dataset, the suggested HOG-based feature extraction and SVM classification method attained an accuracy of 95.4%.
[26]	2020	GAN, U-NET	HIT-MW, ICDAR 2013	The GAN-based model obtained an F-measure of 0.9412 on the HIT-MW dataset, while the previous best approach obtained an F-measure of 0.8912. The GAN-based model outperformed the previous best result of 0.9248 on the ICDAR 2013 dataset, achieving an F-measure of 0.9456.
[28]	2017	Overview of CNN	No dataset used	The writers emphasize the main benefits of CNNs, including their efficacy in computer vision applications, translation invariance, and their capacity to automatically extract features from unprocessed input.
[33]	2018	CNN, RNN (Bi-LSTM)	AOLP (AI-based Online License Plate Recognition) dataset CCPD (Chinese Car License Plate Detection) dataset	The suggested CNN-RNN technique obtained 97.8% accuracy in character recognition and 96.5% accuracy in license plate recognition on the AOLP dataset. The CNN-RNN model obtained a 95.8% accuracy rate for license plate identification and a 97.1% accuracy rate for character recognition on the CCPD dataset.
[34]	2024	CNN, SVM	IAM Dataset	On the IAM dataset, the suggested hybrid technique produced a character recognition accuracy of 94.2%. Compared to employing the CNN or SVM classifiers separately, which produced results of 91.8% and 92.5%, respectively, this is a notable improvement.
[36]	2016	MDLSTM (Multi-dimensional long short-term memory)	IAM Dataset, RIMES Dataset	The model obtained a character error rate (CER) of 4.8% at 150 dpi resolution and 6.6% at 300 dpi resolution on the IAM dataset. The model obtained a CER of 4.8% at 150 dpi resolution and 3.6% at 300 dpi resolution on the RIMES dataset.
[37]	2022	VAN (Vertical Attention Network)	IAM Dataset	On the IAM dataset, the suggested VAN-based method produced a character error rate (CER) of 4.54%. The model achieved equivalent results to ground-truth segmentation approaches and outperformed conventional methods based on explicit line segmentation.

B. Limitations

The system's primary drawback is that it uses preset classes and Romanized identified words for fuzzy-based matching, which could not work as well for texts with complex or non-standard Hindi text [2]. The dependency of the system on a sizable annotated dataset of old handwritten documents, which can be hard to come by and expensive to acquire. When dealing with documents that have significant deterioration or unique layouts that are absent from the training set, the system's performance may also suffer [3].

The requirement for more computational resources is one of the limitations; this could present a problem for systems with little processing capacity [5]. Fuzzy-based matching relies on Romanized identified words, which might not work as well for texts with complex or non-standard Hindi text. More sophisticated text recognition methods designed for the variety of Indian language scripts could enhance performance [9]. The dependency on a sizable, excellent collection of handwritten Tamil papers, might be difficult to come by. Furthermore, documents with extreme deterioration or atypical layouts that are absent from the training set may result in a decline in performance [11].

Processing the full paragraph at once, which results in significant computational and memory requirements is the fundamental drawback due to which, it is difficult to implement in real-world applications where efficiency and speed of processing are essential [12]. Lack of labeled data [17], Sensitivity to the angle of orientation [18], Quality Analysis of Arabic letters [19], Model complexity [20], Multilingual characteristics of many invoices [21], Complexity of segmentation

[22], Intricacy of Handwritten text [23], Over-lapping in text line documents [24], Difficulty in Handling noise and degradation [25], Limited Applicability [26] are also the major drawbacks.

The model MDLSTM must process the full paragraph at once, which results in significant computational and memory requirements. This is the fundamental drawback of this approach. Because of this, it is difficult to implement in real-world applications where efficiency and speed of processing are essential [36].

III. PROPOSED METHODOLOGY FOR WORD EXTRACTION

In this section, OCR methods are used which are based on CNN and RNN to extract text from handwritten document images. Figure 3 illustrates the sequential process of extracting text from an image-based form document. This system predominantly comprises four key steps: Pre-processing, Feature Extraction, Image Segmentation, and Classification. The approach employed for the handwritten character recognizer is outlined in Figure 3.

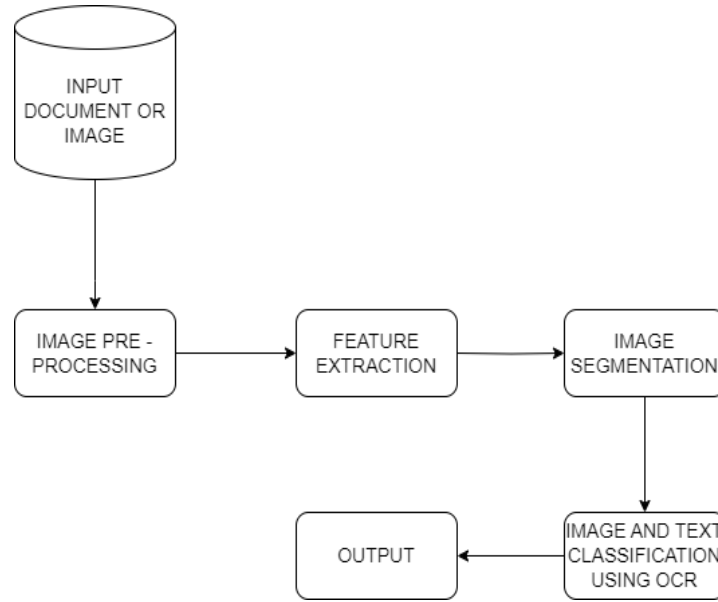


Fig. 3. Proposed Methodology

The following are the stages of the proposed methodology explained:

First, provide the document or picture for the analysis. It could be a picture, a scanned document, or any other kind of visual data that includes both text and pictures.

A. Image Preprocessing

The preparation of handwritten text entails a sequence of operations executed on the data before applying recognition algorithms. This phase tackles dimensionality reduction, inconsistency removal, and data normalization issues, resulting in a dataset better suited for image format data segmentation.

B. Feature Extraction

Extract relevant features from the preprocessed image. Features could include visual elements such as shapes, textures, colors, and patterns. Feature extraction is crucial for capturing meaningful information from the image that can be used for subsequent analysis.

C. Image Segmentation

Divide the image into meaningful segments or regions. Image segmentation helps in identifying distinct objects or areas within the image, which can aid in better understanding and analysis. This step is particularly important for complex images with multiple objects or components. The result of line segmentation is text lines, which are then separated into individual words in the word segmentation process.

D. Image Classification

Apply OCR to extract text from the image. OCR algorithms analyze the image to recognize and convert text elements into machine-readable format. Simultaneously, perform image classification to categorize the visual content of the image based

on the extracted features. This step involves training CNN and RNN models to classify images into predefined categories or classes.

E. Results Analysis

Finally, presents the results of the analysis, which may include the extracted text, segmented image regions, and classifications obtained from both image and text analysis. The output could be in the form of structured data, visual representations, or any other format suitable for the intended application or user requirements.

IV. DATASET DESCRIPTION AND SYSTEM CONFIGURATION

A. Dataset Description

There are a number of segmentation and recognition tasks that make use of handwritten samples, and the most popular of them are the IAM database and the RIMES (Recognition and Indexing of handwritten documents and faxes) database [32]. Handwritten letters, essays, and notes are among the many types of American handwritten literature found here. Some image-processing methods for separating lines and words from handwritten text and extracting them from forms are also part of the IAM and RIMES databases.

Then utilized the IAM and RIMES dataset for model training, and then augmented it with photos of handwritten paragraphs provided by random individuals for testing. Figure 4 depicts the sample of the IAM database, figure 5 depicts the sample of the RIMES database.

B. Training and Testing

For training and testing, the datasets are divided into two ratios: 80:20 and 70:30. 20 percent of the word images in the IAM dataset are used for training and 80 percent for testing, whereas seventy percent of the word images in the RIMES dataset are used for training and thirty percent are used for testing. Both of these percentages are utilized to train the model. Random selection and preprocessing are performed on the handwritten custom images and paragraph images that are included in the IAM dataset.

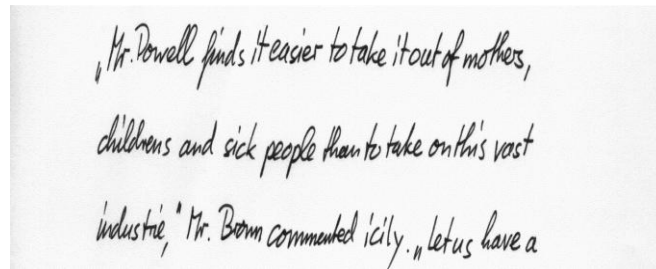


Fig. 4. Sample of IAM Dataset [39]

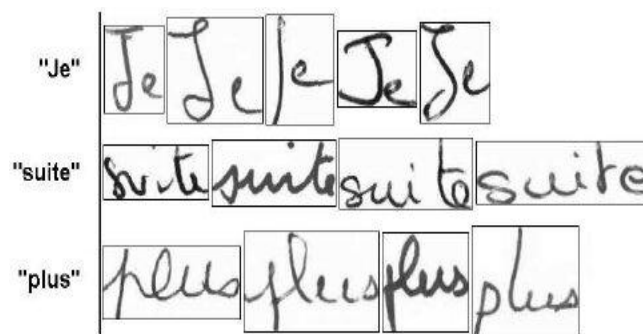


Fig. 5. Sample of RIMES Dataset [40]

Table II. Dataset split in training and testing

Dataset	Level	Training	Testing
RIMES	Paragraph	10,193	778
I AM	Paragraph	6,482	976

C. System Configuration

Hardware Requirements: 16 GB of RAM as a minimum to handle big datasets and models, Ample storage space (such as a 1 TB SSD) to store trained models and datasets.

Software Requirements: Operating System - Windows 10/11, Python 3.7 or later is required to program for deep learning. Using PyTorch 1. x or TensorFlow 2. x to create and train deep learning models, OpenCV for image manipulation and preprocessing, Natural Language Toolkit (NLTK) for text analysis and preprocessing

V. RESULTS AND DISCUSSION

According to an analysis of about 100 research papers Figure 5(a) shows the findings of scripts used widely for text and information extraction processes, Figure 5(b) shows the recognition levels, be it character, word, line, or paragraph, Figure 5(c) depicts the data augmentation approaches widely used in the field of text and information extraction and lastly, Figure 5(d) shows the deep learning models used.

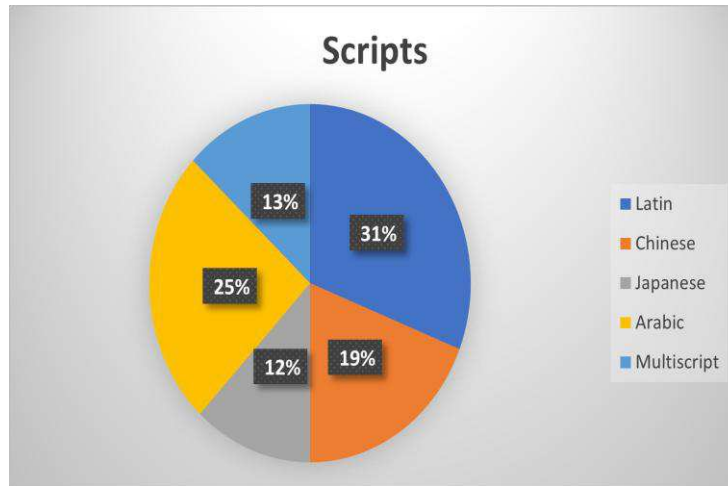


Fig. .4(a) Various corpora studied according to script.

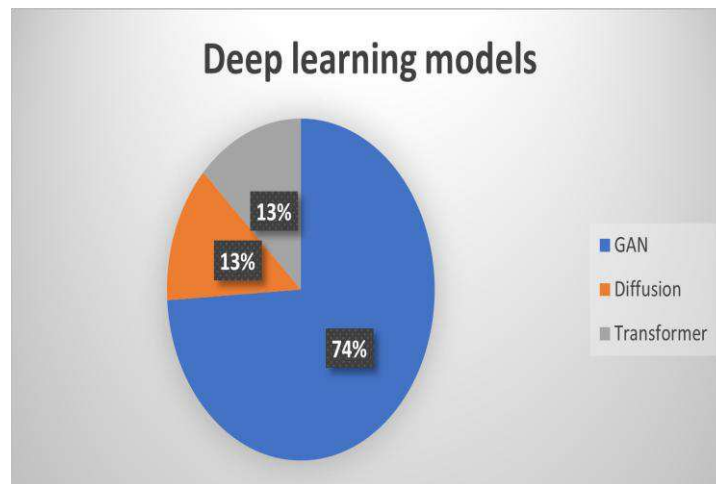


Fig. .4(b) Proportion of recognition levels.

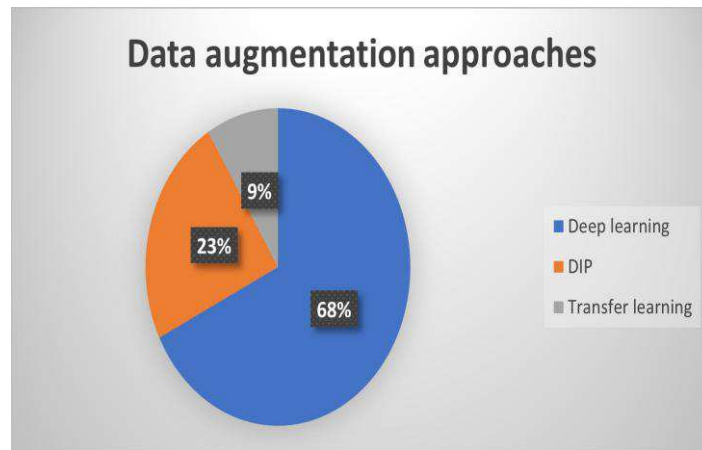


Fig. .4(c) Data augmentation approaches used.

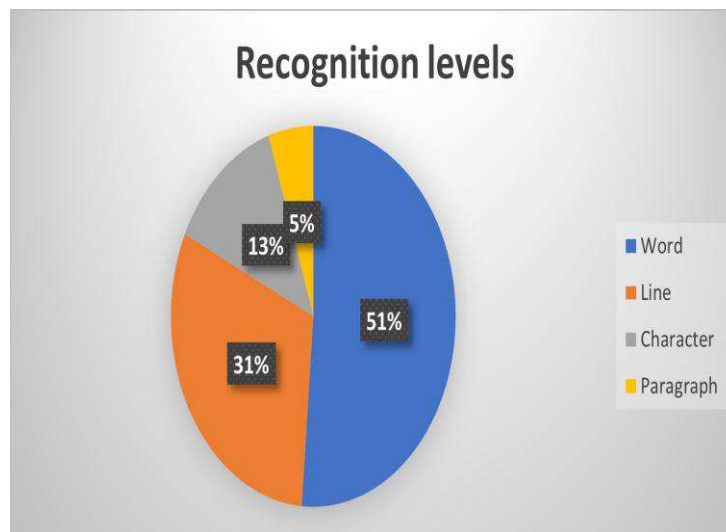


Fig. .4(d). Deep learning models used

The pre-processing implementation consists of two primary stages: Line segmentation and word segmentation, utilizing the OpenCV library.

A. Line Segmentation

Line segmentation involves initial pre-processing of the input image, converting it to grayscale as depicted in Figure 7. Subsequently, the grayscale image is transformed into an inverse binary image, as demonstrated in Figure 8. The resulting inverted binary image undergoes dilation, illustrated in Figure 9. Contours are then identified within the binary image and enclosed with bounding boxes, as shown in Figure 10. These segmented lines are subsequently saved as individual line images, depicted in Figure 11.

Handwriting has remained one of the most frequently occurring patterns that we come across in everyday life. Handwriting offers a number of interesting text classification problems including handwriting recognition, writer identification, signature verification, writer demographics classification and script recognition, etc.

Fig. .7 Gray Scale Image

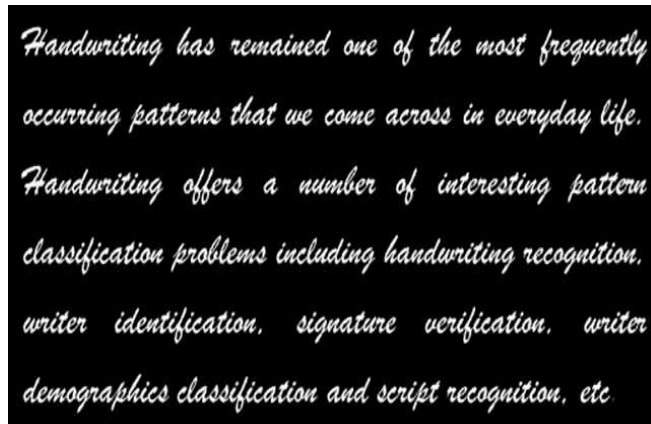


Fig. 8. Inverted Binary Image



Fig. 9. Dilated Image

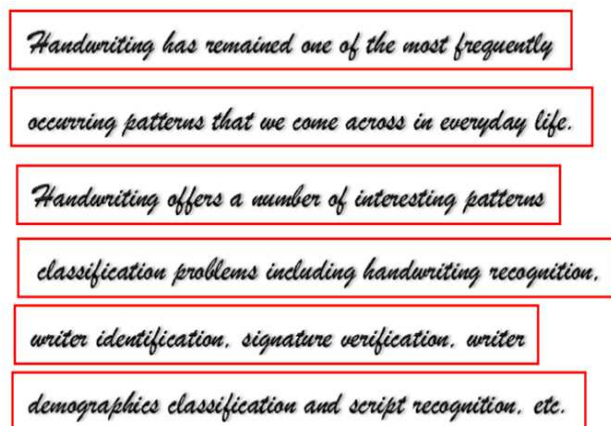


Fig. 10. Bounding Boxes over lines



Fig. 11. Lines Images

B. Word Segmentation

As seen in Figure 12, the line image is first pre-processed before the filter kernel is executed. In Figure 13, they see the filtered picture being dilated. Figure 14 shows the inverted binary image that is the result. Figure 15 shows the final result of storing the words as individual word pictures.



Fig. 12. Filtered line over image



Fig. 13. Dilated line image



Fig. 14. Inverted line image



Fig. 15. Segmented Word Images

C. Performance analysis

An efficient method is used to develop the offline handwritten character extraction system to enhance its functionality at each stage of the process, beginning with the pre-processing stage and continuing through training and testing.

D. Text Extraction

The quality of the text extraction is evaluated based on the CER and the WER, respectively. At the paragraph level, both of

them are estimated using the Levenshtein distance between the text that is considered to be the ground truth and the text that is forecast. Tables III and IV show the paragraph level on the IAM and RIMES datasets.

TABLE III The paragraph level (for 100 paragraphs) on the IAM dataset

Architecture	CER Testing	WER Training	CER Testing	WER Training
Proposed model with RNN	5.3	16.42	7.4	17.62
Proposed model with CNN	6.2	17.91	8.1	18.45

TABLE IV The paragraph level on the RIMES dataset

Architecture	CER Testing	WER Training	CER Testing	WER Training
Proposed model with RNN	5.8	17.43	8.4	19.62
Proposed model with CNN	7.5	18.03	9.3	20.37

E. Evaluation matrix

We measure the success of the proposed method using the following key performance indicators:

- T_P (True positive): the number of borrowers correctly classified as {approved}.
- T_N (True negative): the number of borrowers correctly classified as {rejected}.
- F_N (False negative): the number of borrowers incorrectly classified as {rejected}.
- F_P (False positive): the number of borrowers incorrectly classified as {approved}.

To evaluate the suggested technique and to make comparisons to others, the authors compute its accuracy, precision, recall, and f1-score [35]. The formulas for these are as follows:

$$Accuracy = \frac{T_P + T_N}{T_{Total_Images}} \times 100$$

$$Precision = \frac{T_P}{T_P + F_P}$$

$$Recall = \left(\frac{T_P}{T_P + F_N} \right)$$

$$F1 - score = \frac{2 \times Precision \times Recall}{Precision + Recall}$$

Two experiments are conducted:

(1) a performance evaluation of each model in this study when the dataset is split into a training set and a test set in an 8:2 ratio.

(2) This research compares the performance of each model when the dataset is split into a training set and a test set in a 7:3 ratio. Tables V and VI indicate the confusion matrix of CNN and RNN in the training and testing dataset.

TABLE V For IAM data 8:2

	Positive	Negative
--	-----------------	-----------------

	RNN	
True	TP = 428	TN = 264
False	FP = 51	FN = 49
	CNN	
True	TP = 274	TN = 193
False	FP = 37	FN = 19

TABLE VI For RIMES data 7:3 word

	Positive	Negative
	RNN	
True	TP = 391	TN = 184
False	FP = 58	FN = 56
	CNN	
True	TP = 193	TN = 127
False	FP = 35	FN = 17

F. Analysis of Result

Tables VII and VIII provide the results of the performance evaluations of the prediction models applied to the training dataset and the testing dataset, respectively. The tables provide an evaluation of the accuracy of category prediction as well as the discriminant ability of the classifiers that were included.

“The True Positive Classification Rate (TP), the False Positive Classification Rate (FP), the True Negative Classification Rate (TN), and the False Negative Classification Rate (FN) are shown for CNN and RNN. Figures 15 and 16 show the graph of different classifiers for IAM data and RIMES data.”

TABLE VII Performance indicators of different classifiers for the IAM dataset

Techniques Used	Accuracy	Precision	Recall	F1-score
RNN	87.3%	89.3%	89.7%	89.4%
CNN	89.3%	88.1%	93.5%	99.01%

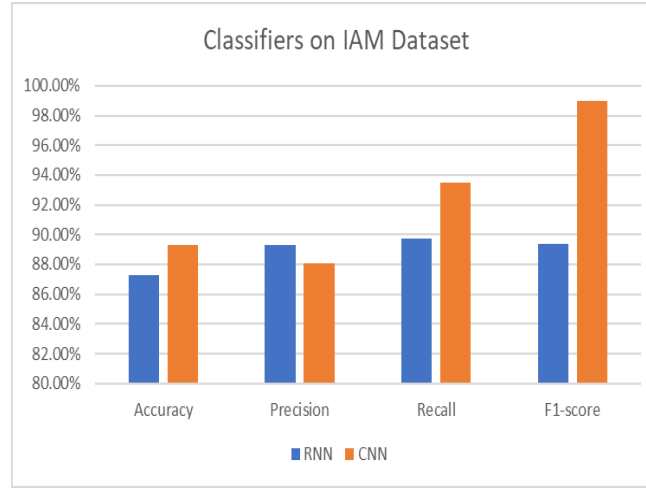


Fig. 15 Graph of Different Classifiers for IAM dataset

TABLE VIII Performance indicators of different classifiers for the RIMES dataset

Techniques Used	Accuracy	Precision	Recall	F1-score
RNN	83.4%	87%	87.4%	87.1%
CNN	86%	84.6%	91.9%	88%

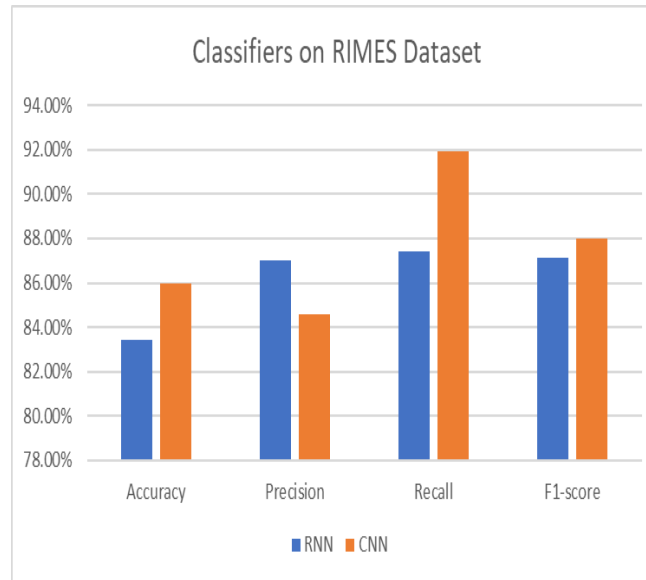


Fig. 16 Graph of Different Classifiers for RIMES dataset

G. Comparative Analysis

The findings of the proposed model are shown in Tables IX and X, respectively, after being compared to the techniques that are considered to be state-of-the-art for the RIMES and IAM datasets. The fact that we can get comparable results on all two datasets, each of which has its intricacies, without making any hyperparameter adjustments is something that can happen. The results shown here involve pretraining of the model on images; however, the model could have been trained without needing to undergo pretraining.

TABLE IX Analysis of RIMES dataset paragraph-level performance comparing the suggested methods to state-of-the-art techniques

Architecture	CER Testing	WER Testing	CER Training	WER Training
CNN+MDL STM [36]	2.5	12.0	2.9	12.6
VAN [37]	1.74	8.72	1.90	8.83
RNN	5.8	17.43	8.4	19.62
CNN	7.5	18.03	9.3	20.37

Figure 17 depicts the comparison graph of the proposed method with another existing method which is based on the RIMES dataset as shown below. The RNN method attained a CER testing of 5.8 and WER training of 17.43 while the CNN method obtained a CER testing value of 7.5 and WER training values of 18.03 as shown in Table IX.

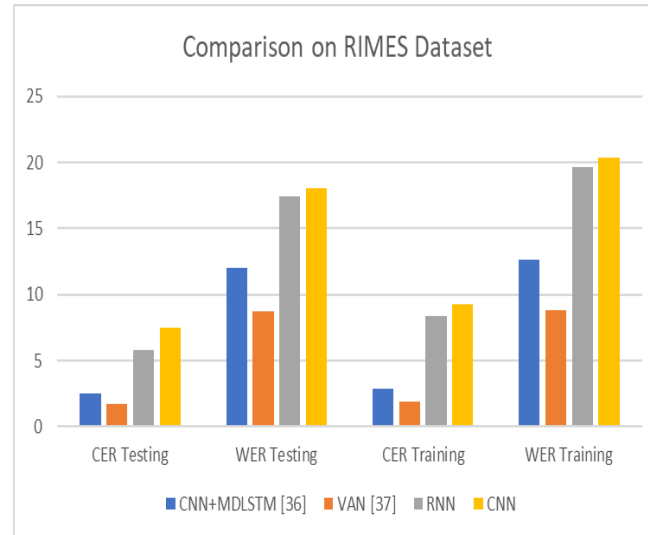


Fig. 17 Comparison graph of the proposed method on the RIMES dataset

TABLE X Analysis of IAM dataset paragraph-level (for 100 paragraphs) performance comparing the suggested methods to state-of-the-art techniques

Architecture	CER testing	WER training	CER testing	WER training
CNN+MDL STM [36]	4.9	17.1	7.9	24.6
VAN [37]	3.04	12.69	4.32	16.24
Proposed model with RNN	5.3	16.42	7.4	17.62
Proposed model with CNN	6.2	17.91	8.1	18.45

Figure 18 depicts the comparison graph of the proposed method with another existing method which is based on the IAM dataset as shown below. The RNN method attained a CER testing of 5.3 and WER training of 16.42 while the CNN method obtained a CER testing value of 6.2 and WER training values of 17.91 as shown in Table X.

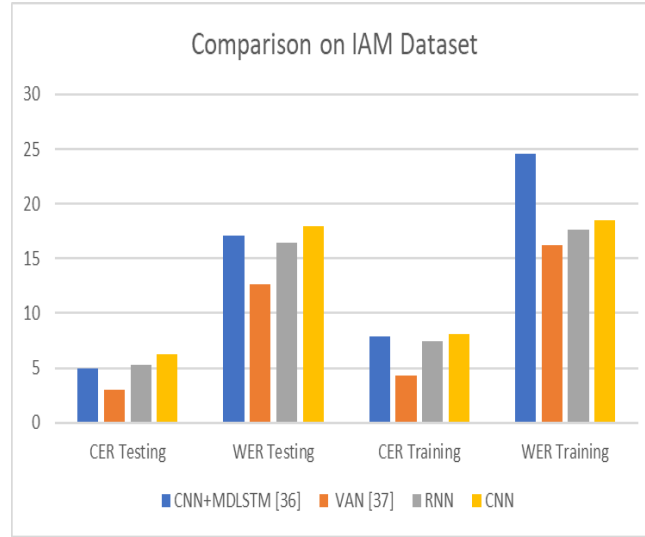


Fig. 18 Comparison graph of the proposed method on the IAM dataset

VI. CONCLUSION AND FUTURE SCOPE

Text extraction from handwritten document images is a crucial task with wide-ranging applications, from digitizing historical documents to aiding visually impaired individuals. Through advancements in OCR techniques by combining CNNs and RNNs, significant progress has been made in accurately extracting text from handwritten images. OCR, or Optical Character Recognition, is a technique that allows for the transformation of scanned pictures of printed or handwritten text into characters that can be read by machines. These characters may then be modified, searched, and analyzed electronically. This software is specifically designed for the purpose of recognizing and interpreting text written in the English language. OCR facilitates the conversion process, making it more efficient and less sensitive to the original material. Users have the ability to modify, structure, and search the document, enhancing its user-friendliness. After OCR transforms a handwritten document into Word and text format, customers have the option to save their papers either on their local computers or on our server, depending on their requirements. The quality of text recognition was evaluated using the CER and WER. The research included two datasets, IAM and RIMES, which were partitioned into separate sets for training and testing purposes. By use of this segmentation, the metrics of accuracy, precision, and recall were established. The experimental results demonstrate that the CNN technique attained superior accuracy rates on both datasets, achieving 89.3% and 86% respectively.

Enhancing text extraction algorithms' resilience and accuracy should be the focus of future research. This might be achieved by exploring multimodal methods and synthetic data creation strategies.

Data availability

The dataset used are IAM and RIMES, which are available at <https://fki.tic.heia-fr.ch/databases/download-the-iam-handwriting-database> (for IAM database) and http://www.a2ialab.com/doku.php?id=rimes_database:start (for RIMES database)

Authors Contribution

D.S. and S.P. conceived the study. D.S. contributed to the study design. Data analysis and figures were performed by S.P.. Data collection was performed by S.P. Data curation was performed by D.S.. The first draft of the manuscript was written by D.S., S.P., All authors read and approved the final manuscript.

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Declarations

Competing interests

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